The Hidden Costs of Family Bonds: Labor Market and Health Outcomes of Dual Caregiving in China*

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Abstract

With population aging and changes in family structures, middle-aged women increasingly face dual caregiving responsibilities of simultaneously caring for parents and grandchildren. In China, this "sandwich generation" phenomenon is particularly pronounced due to traditional family values and limited formal care services. This study uses China Health and Retirement Longitudinal Study (CHARLS) data from 2011-2018 to analyze the impact of dual caregiving on middle-aged women's labor supply and health status. The study employs a progressive analytical strategy. I begin with descriptive analysis of caregiving exposure patterns among women aged 45-64. Then I examine responses to exogenous shocks, including parental death, health deterioration, and grandchild birth. Finally, I conduct causal identification using instrumental variable methods with the number of living parents and grandchildren as instruments. In the causal section, I separate caregiving type from caregiving intensity to distinguish between whether one undertakes dual caregiving versus the degree of caregiving burden. The main findings include three key patterns. First, caregiving intensity rather than caregiving type drives most negative outcomes. High-intensity caregiving leads to significant income reductions and mental health deterioration. Second, dual caregiving itself has limited independent effects but shows a positive impact on income, suggesting complex resource allocation mechanisms within families. Third, dual caregiving does not amplify the negative effects of high-intensity caregiving. Heterogeneity analysis shows important group differences. Urban women bear economic shocks approximately twice those of rural women, reflecting differences in labor market formalization. Women with lower education levels face greater economic and health risks. These findings may provide empirical evidence for targeted caregiving support policies that focus on reducing caregiving intensity rather than redistributing caregiving types.

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1 Introduction

China is experiencing a rapid demographic transition. By the end of 2023, the population aged 65 and above reached 216.76 million, accounting for 15.4% of the total population[1]. With the aging trend and increased life expectancy, four-generation families (great-grandparents, grandparents, parents, and children coexisting) are becoming increasingly common. Against this backdrop, a growing number of middle-aged individuals, particularly those aged 45-64 in the later stages of their careers, face the dual pressure of caring for elderly parents and young grandchildren, forming what is known as the "sandwich generation." Data from the 2011 China Health and Retirement Longitudinal Study (CHARLS) show that 32.7% of middle-aged adults aged 45-64 have both at least one living parent or parent-in-law and young grandchildren. Among this sandwich generation, 58% provide care for grandchildren, 23% provide care for parents, and approximately 15% simultaneously care for both parents and grandchildren (Falkingham et al., 2020).

While the "sandwich generation" phenomenon exists across countries, Chinese women in this group face caregiving burdens that differ significantly from their Western counterparts due to China's unique socio-cultural and institutional context. First, traditional values emphasizing filial piety and family obligations make informal care, particularly care provided by women, a moral duty rather than simply a choice. As a result, institutional care faces greater social resistance compared to Western societies. According to National Health Commission data, as of 2021, approximately 90% of Chinese elderly chose home-based care, 7% relied on community support, and only 3% opted for professional care institutions. Similarly, the formal childcare system shows significant gaps, especially for children under three years old. The National Population and Family Dynamic Monitoring Survey shows that in 2019, only 4.3% of children under three were enrolled in formal childcare facilities, making grandparental care an essential supplement for young parents balancing work and family responsibilities.

These structural and cultural factors not only affect caregiving outcomes but fundamentally shape caregiving decisions and caregiver behavior. Compared to Western societies, Chinese middle-aged adults, particularly women, face greater social pressure and fewer alternatives in caregiving decisions. Choosing not to care for elderly parents and grandchildren personally may result in significant social stigma, while limited public resources make caregiving almost

[1] Data source: 2023 National Development Report on Aging Care.

inevitable. These multiple caregiving responsibilities may impact their career development, income levels, and physical and mental health. This intergenerational caregiving pressure is expected to intensify further, particularly as delayed retirement policies are gradually implemented.

Therefore, this study aims to answer a key question: within China's unique socio-cultural and institutional context, how does the dual caregiving burden of women in the "sandwich generation" affect their labor market participation and health status? Specifically, this study focuses on three questions: (1) Does potential dual caregiving impose greater caregiving pressure on caregivers, as measured by caregiving hours? (2) When facing exogenous shocks, how do the response patterns differ among different types of potential caregivers? (3) What are the actual impacts of dual caregiving, and what drives these impacts?

In general, the research findings indicate three key patterns. First, while potential dual caregivers do not always bear the highest caregiving pressure, they still face considerable stress with more complex caregiving tasks. Second, different types of caregivers respond differently to the same shocks, primarily in terms of the magnitude and absolute values of changes in labor and health outcomes. Third, dual caregiving behavior itself does not significantly affect caregivers' working hours, physical health, or mental health, and may even promote income increases. The main factors contributing to respondents' income loss and mental health deterioration are high-intensity caregiving. Moreover, dual caregiving does not amplify the negative effects of high-intensity caregiving. In other words, the core issue lies in intensity rather than the type of caregiving: whether single or dual caregiving, once reaching high-intensity levels, both significantly harm caregivers' mental health and income. High-intensity caregiving is the primary cause of various negative outcomes, and this effect is more pronounced among urban samples and female samples with lower educational levels.

The potential contribution of this paper to the literature is in three ways. First, I employ instrumental variable methods to address the endogeneity problem in caregiving decisions, an issue that has often been overlooked in previous research. Second, I conduct a joint analysis of labor market and health outcomes to provide a more comprehensive picture of dual caregiving impacts. Third, I separate caregiving types from caregiving intensity in the analysis, finding that what truly brings significant impacts is not the caregiving type itself, but caregiving intensity. This finding challenges previous literature that focused solely on caregiving types and provides new insights for understanding the mechanisms of caregiving burden.

2 Literature Review

2.1 Elderly Caregiving: Impacts on Labor and Health Outcomes

Research on the economic and health consequences of providing parental care is relatively well-developed. Most studies show consistent results: elder care negatively affects employment and working hours (Jiang and Zhao, 2009; Nguyen and Connelly, 2014), particularly for women. In the Chinese context, Wu et al. (2017) found that family elder care reduces female labor force participation by 4.5% and weekly working hours by 2.7 hours, based on China Health and Nutrition Survey data from 1993-2011. This study also found significant effect heterogeneity: rural women are more affected than urban women; co-residence with parents has a greater impact on women's labor participation; and when there are more siblings, the impact of family elder care on women's employment is reduced. Similarly, in terms of health, providing elder care is significantly associated with deterioration in caregivers' mental and physical health (Converso et al., 2020), especially under high-intensity caregiving conditions.

In methodological terms, instrumental variable (IV) methods are commonly used to address endogeneity issues in this type of research (Crespo and Mira, 2014; Jiang and Zhao, 2009). Fevang et al. (2012) found in Norwegian registry data that parental death events mainly affect adult children's labor supply through caregiving demand, providing a theoretical foundation for instrument variable selection in this study.

2.2 Grandchild Caregiving: Impacts on Labor and Health Outcomes

Research on how grandparental caregiving affects elderly caregivers themselves mainly focuses on health outcomes, with analysis of labor market impacts gradually increasing but still relatively limited.

Regarding health impacts, existing research presents diverse viewpoints. Eibich and Zai (2022) analyzed European data and found that providing grandchild care is associated with declines in grandparents' physical functioning and subjective health. In contrast, Di Gessa et al. (2016), also using European SHARE data, reached different conclusions, emphasizing the importance of caregiving intensity: they argue that moderate grandchild care may provide grandparents with psychological satisfaction and social participation benefits. In the Chinese context, Chen and Liu (2012) found a nonlinear relationship between caregiving intensity and health impacts: moderate caregiving may be beneficial, but high-intensity caregiving may harm health, indicating that cultural factors and caregiving intensity play important roles in impact mechanisms.

Regarding labor market impacts, international research presents diverse conclusions. One type of study shows that caring for grandchildren significantly reduces elderly labor participation rates and working hours (Zanella, 2017), with elderly women's labor supply being particularly affected (Wang and Marcotte, 2007; Rupert and Zanella, 2018). Another type of research finds that under certain circumstances, caring for grandchildren may actually promote elderly labor participation and increase working hours, especially for married grandfathers (Wang and Marcotte, 2007). Married grandfathers are more inclined to employment, and when grandfathers co-reside with grandchildren, their working hours increase (Ho, 2015). In the Chinese background, Long and Yuan (2019) analyzed CHARLS data and found that providing intergenerational care significantly reduces middle-aged and elderly labor participation rates, with particularly pronounced effects on middle-aged women. Wang et al. (2021) research on rural samples further confirmed this negative impact, especially among grandmother groups aged 61-70.

2.3 Dual Caregiving: Existing Evidence and Research Gaps

Research on dual caregiving impacts mainly focuses on health outcomes. Most studies show that simultaneously caring for parents and the younger generation may lead to depression (Brenna, 2021; Cheng and Santos-Lozada, 2024), occupational burnout, and physical health deterioration (Do et al., 2014). In the Chinese context, research on dual caregiving impacts remains limited and methodologically relatively simple. However, existing studies have revealed unique patterns: unlike international research findings, Chinese studies indicate that such caregiving may harm physical health but benefit mental health, presenting culturally specific bidirectional effects (Shi, 2024; Hsu et al., 2023).

Despite providing valuable insights, existing research still has several significant research gaps: first, few studies simultaneously examine the comprehensive impact of caregiving on health and labor markets, limiting the overall assessment of caregiving's economic and welfare consequences; second, most research focuses on single caregiving responsibilities, with insufficient attention to the "sandwich generation" that simultaneously bears dual caregiving responsibilities for both parents and grandchildren; third, existing dual caregiving research is mainly limited to binary analysis of caregiving status (whether one is a dual caregiver), failing to effectively distinguish between the different roles of caregiving types and caregiving intensity, and ignoring that caregiving intensity may be a key factor determining caregiving impacts; fourth, methodologically, endogeneity issues in dual caregiving research remain insufficiently addressed, lacking effective causal identification strategies.

These gaps limit our understanding of how caregiving behaviors affect the welfare of different populations. Accordingly, this study aims to fill these research gaps through a rigorous methodological design and comprehensive outcome examination.

3 Data

3.1 Data Sources

This study uses three waves of the China Health and Retirement Longitudinal Study (CHARLS) from 2011, 2013, and 2018. CHARLS is a nationally representative survey that covers approximately 10000 households and 17000 individuals aged 45 and above across 150 counties and 450 villages in each wave. The three waves include over 25000 total observations. The strength of CHARLS lies in its comprehensive collection of respondents' employment and health information, alongside detailed data on intergenerational interactions, including care provision and receipt between family members. For the questions focus on caregiving arrangements, the survey employs a two-step design. It first asks whether respondents have caregiving responsibilities, separately for grandchildren and parents. Then it inquires about the specific time spent caring for each individual. This approach enables precise measurement of actual caregiving hours rather than relying solely on binary caregiving status. Such detailed measurement is crucial for analyzing caregiving intensity effects.

However, data on care hours for parents are completely missing from the 2015 and 2020 surveys. This limitation leads to excluding these waves from the analysis. The analytical sample focuses on women aged 45-64, a demographic choice driven by two key considerations. First, this age group represents the classic "sandwich generation" that simultaneously faces pressures from aging parents and young grandchildren. Second, women typically shoulder disproportionately higher caregiving burdens compared to men and face more complex trade-offs between work and family responsibilities. After applying the sample selection criteria, the final analytical sample comprises 17087 observations across 8240 households over the three survey waves.

3.2 Variables Construction

3.2.1 Caregiving Variables

To separate the effects of dual caregiving responsibilities and caregiving intensity, this study constructs two distinct dummy variables. As described above, CHARLS data contain binary responses indicating whether each respondent provides care for parents and grandchildren, enabling the construction of a dummy variable distinguishing actual dual caregivers from non-dual caregivers. The dual caregiving dummy variable equals 1 if respondents simultaneously provide care for both parents and grandchildren, and 0 otherwise.

Additionally, the questionnaire also contains data on caregiving time for parents and grandchildren separately, including the number of weeks per year and hours per week spent on caregiving. Based on this information, total annual caregiving hours can be constructed. Total caregiving time is defined as the sum of parental care hours and grandchild care hours. For constructing the caregiving intensity variable, this study follows common practice in the literature (Jacobs et al., 2017; Kolodziej et al., 2022) by setting 20 hours per week as the threshold for high-intensity caregiving (1040 hours annually). For respondents with caregiving behavior, the high-intensity caregiving dummy variable equals 1 if their total caregiving time exceeds this threshold, and 0 otherwise. This specification allows us to distinguish between caregiving type (whether one undertakes dual caregiving) and caregiving intensity (the degree of caregiving burden), thereby identifying the key factors that truly drive caregiving effects.

3.2.2 Outcome Variables

This study examines two main categories of outcome variables. First, labor market performance, specifically measured by total annual working hours and total annual income. Second, health status, measured by the number of chronic diseases and depressive symptoms as indicators of physical and mental health, respectively:

Chronic disease count. CHARLS survey asks respondents about their diagnosis status for 14 common chronic conditions: hypertension, high (or low) blood lipids, diabetes, cancer, chronic lung disease, liver disease, heart disease, stroke, kidney disease, digestive system disease, mental illness, memory-related disease, arthritis or rheumatism, and asthma. Each condition is scored as 1 if the respondent reports having the disease, and the total number of chronic diseases serves as a proxy indicator for respondents' overall chronic disease burden.

Depressive symptoms. Cong and Silverstein (2008) adapted the Center for Epidemiologic Studies Depression Scale (CES-D-10) for Chinese elderly populations, demonstrating good reliability and validity in Chinese samples. The scale contains 10 items, with CHARLS modifying the original 1-4 point scoring system to a 4-point scale: "0 (rarely or never), 1 (not much), 2 (sometimes), 3 (most of the time)." Total scores range from 0-30, with higher scores indicating more severe depressive symptoms. CHARLS uses this scale to measure respondents' depressive symptoms. In this study, the scale demonstrates good internal consistency reliability with a Cronbach's α coefficient of 0.771.

3.2.3 Instrumental Variables

Whether individuals undertake dual caregiving responsibilities and the intensity of care they provide are likely related to personal characteristics of caregivers, potentially creating endogeneity concerns. To address this issue, this study employs instrumental variable methods. While instrumental variable approaches are commonly used in single caregiving studies, their application in dual caregiving research remains limited. To fill this methodological gap, this study applies instrumental variables to both dummy variables: whether one provides dual caregiving and whether one provides high-intensity caregiving. The specific instrumental variables are the number of living parents, the number of grandchildren under 16 years old, and their interaction.

4 Descriptive Analysis

4.1 Static Analysis: Distribution of Caregiving Burden

Figure 1 shows the distribution of caregiving responsibilities among women aged 45-64 in CHARLS 2013. Potential dual caregivers (those who simultaneously have parents and grandchildren) constitute a substantial proportion, reaching 32.6% (255 individuals). Additionally, actual dual caregivers identified through the survey account for 3.7% of the total sample (59 individuals). Therefore, a total of 36.3% of women face dual caregiving demands, indicating that dual caregiving is not an isolated phenomenon but rather a widespread challenge confronting contemporary Chinese families.

The sample of actual caregivers suffers from significant endogeneity concerns, as caregiving decisions are often influenced by unobservable factors such as individual health status, labor

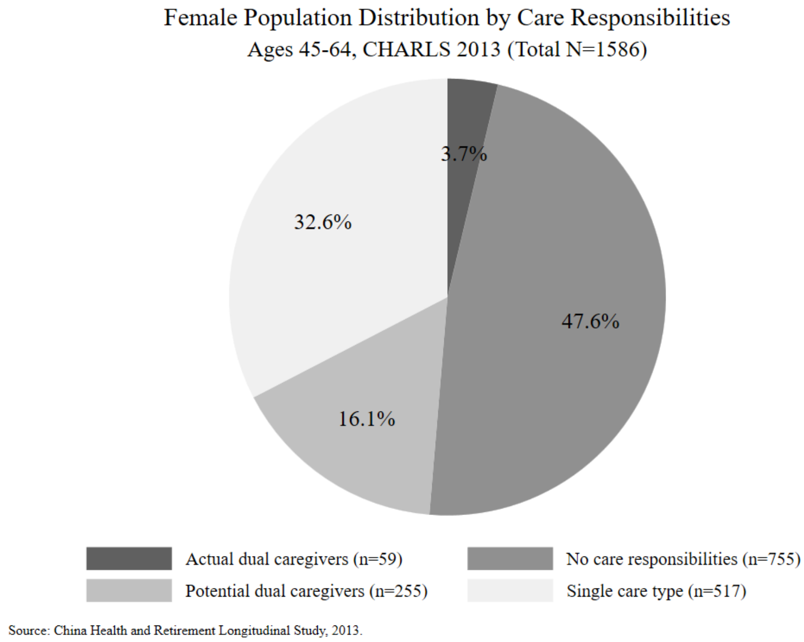

Figure 1: Female Population Distribution by Care Responsibilities

market performance, and household economic conditions. To obtain more credible research results, this study employs a progressive analytical strategy: first establishing a descriptive evidence base through static analysis and exogenous shock analysis from a "caregiving exposure" perspective, then conducting in-depth exploration of causal mechanisms through instrumental variable analysis based on "actual caregiving behavior."

In the static and dynamic analyses, this study adopts the concept of "caregiving exposure" to define the research sample. This approach identifies women with parents and/or grandchildren in need of care as potential caregivers. This exposure status possesses greater exogeneity relative to actual caregiving behavior, providing initial and credible evidence regarding whether dual caregiving pressure exists.

To address the descriptive question of whether potential dual caregiving exposure creates greater caregiving pressure for women, I divide the sample into four groups: the dual exposure group (simultaneously facing both parental and grandchild care needs), the parental care exposure group (facing only parental care needs), the grandchild care exposure group (facing only grandchild care needs), and a no-exposure control group. By comparing the distribution of actual caregiving time investments across different exposure groups, this study identifies whether dual exposure indeed creates greater caregiving pressure and more complex caregiving situations.

Figure 2 shows the distribution of caregiving pressure, measured by care hours, across dif-

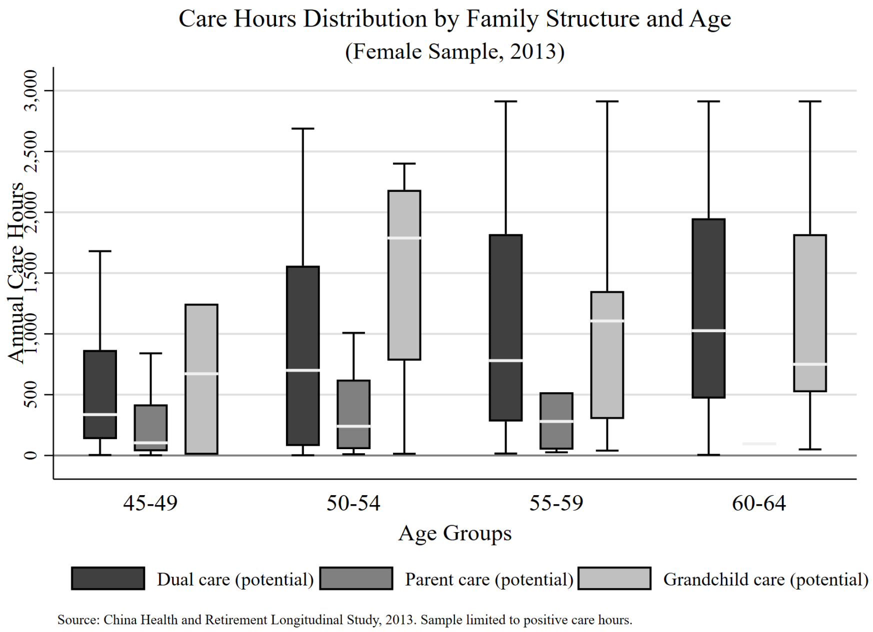

Figure 2: Care Hours Distribution by Family Structure and Age

ferent types of caregiving exposure and age groups. The analysis shows three important characteristics of potential dual caregiving exposure.

First, potential dual caregivers bear significant caregiving time burdens. Although single-exposure caregivers may show higher median care hours in some age groups, dual-exposure caregivers maintain consistently high levels of care investment across all age groups. Moreover, their caregiving hours show a clear upward trend with age throughout the study period, indicating sustained and intensifying caregiving pressure.

Second, dual caregiving exposure demonstrates greater variability in caregiving time. This characteristic is clear across all age groups, indicating substantial differences in caregiving intensity within this group compared to single-exposure caregivers. This finding highlights the importance of distinguishing between dual caregiving behavior and high-intensity caregiving in subsequent analyses.

Finally, dual caregiving exposure creates more complex and diverse caregiving situations. While dual caregivers do not necessarily show the highest upper limits of care hours across all groups, dual caregiving requires simultaneously coordinating care needs for different age groups. This may involve more complex caregiving situations and more diverse caregiving pressures than single caregiving.

Therefore, when facing the same exogenous shocks, dual caregiving exposure may require more complex coping strategies and trade-off mechanisms. These different response patterns may lead to different impacts on the labor market and health outcomes compared to single caregiving exposure. To understand this mechanism in depth, I need to further address the question: when facing exogenous shocks, do different types of potential caregivers show different response patterns, and what are these differences?

4.2 Dynamic Analysis: Response to Exogenous Shocks

The static analysis shows three important characteristics of dual caregiving exposure: significant caregiving time burdens, greater variability in caregiving arrangements, and more complex and diverse caregiving situations. These findings provide initial evidence that dual-exposure caregivers may face unique pressures. However, analysis based solely on cross-sectional data has two important limitations. First, it remains unclear whether the high time burden characteristics of dual caregiving exposure persist across different periods. Second, and more critically, whether dual-exposure caregivers show different response patterns compared to single-exposure caregivers when facing external shocks still needs verification.

To explore these questions in depth, this study adopts a shock analysis approach. I track individuals whose caregiving exposure types remain unchanged during the observation period and examine their behavioral changes in caregiving allocation, labor supply, and health status when facing exogenous shocks. The core logic of this design is as follows: if dual caregiving exposure caregivers show similar response patterns to single-exposure caregivers when facing shocks, this suggests that exposure type itself is not a key factor affecting change. However, if dual-exposure caregivers show stronger labor supply adjustments, larger reconfigurations of caregiving time, or more obvious health deterioration, this would confirm the unique challenges and constraints brought by the overlap of dual responsibilities.

The study uses CHARLS panel data from 2011-2013 and 2013-2018, analyzing three types of relatively exogenous shocks: parental death, parental health deterioration, and grandchild birth. The selection of these three shocks is based on the following considerations. First, these events are largely exogenous and not directly influenced by individual caregiving decisions, helping to reduce endogeneity issues. Second, each shock directly affects respondents' caregiving needs for one generation, providing an ideal context for observing the reallocation of caregiving time between generations and testing response differences across exposure types. Finally, these three shocks cover different scenarios, including reduced caregiving needs, intensity changes, and increased demands, enabling comprehensive testing of the uniqueness of dual caregiving exposure.

Specifically, this study uses descriptive analysis to examine: (1) when facing the same shocks, whether dual caregiving exposure caregivers show different changes in caregiving time compared to single-exposure caregivers; (2) whether dual-exposure caregivers show different changes in labor and health outcomes compared to single-exposure caregivers.

4.2.1 Parental Death Shock

The parental death shock is constructed based on changes in the survival status of respondents' parents (including parents and parents-in-law). Parental death shock is defined as an event where respondents lose at least one parent during at least one observation period. As a relatively sudden exogenous event, the timing of its occurrence has no obvious endogenous relationship with individual labor market performance and other caregiving arrangements, providing a relatively credible basis for causal identification.

Among the 4305 women in the analysis sample who provided complete parental survival information, a total of 1818 people (42.23%) experienced parental death events during the tracking period. Under the conditions requiring stable family structure during the observation period and actual caregiving behavior, I finally obtained 161 valid analysis samples, including 51 dual-exposure caregivers (with grandchildren) and 110 single-exposure caregivers (without grandchildren). Although the sample size is relatively small, the unchanged family structure during the observation period allows us to systematically compare the different response patterns of different caregiving exposure groups to the same exogenous shock while controlling for other factors.

The reallocation of caregiving time shows the unique high-intensity caregiving pressure of dual-exposure caregivers. Figure 3 shows the impact of parental death shock on caregiving time across different caregiving exposure types. After the shock occurs, both caregiving exposure groups show similar caregiving time adjustment patterns, reducing care for parents. In terms of relative changes, both dual-exposure caregivers' reduction in parental caregiving time (approximately 36.3%) and single-exposure caregivers (approximately 23.2%) show significant declines, indicating that parental death shock has important impacts on different caregiving exposure arrangements. The dual-exposure group not only shows larger absolute reductions (1371 hours vs 161 hours), but also simultaneously reduces caregiving time for grandchildren

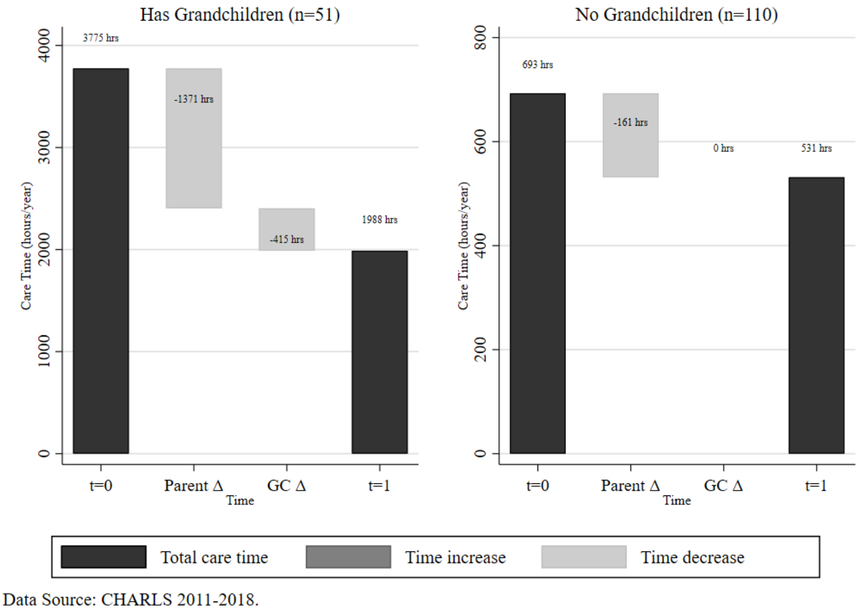

Figure 3: Caregiving Time Reallocation Following Parent Death Shock

(415 hours). However, this group still maintains higher caregiving time (1988 hours) even after experiencing parental death shock. This is essentially because dual-exposure caregivers bear significantly higher baseline caregiving burdens (3775 hours vs 693 hours). Therefore, from the perspective of caregiving time changes, the relative changes of both types of exposure caregivers are significant, but due to differences in initial caregiving time, there are still considerable differences in absolute caregiving time values after the shock.

Similar to caregiving time response patterns, both caregiving exposure groups show patterns that are both similar and different in welfare outcomes. As shown in Figure 4, in terms of change trends, both groups show similar patterns in working hours, physical health, and mental health: working hours all decline, physical health all deteriorate (increased number of diseases), and mental health shows slow deterioration but overall limited change. However, there are significant differences in the magnitude of changes and absolute levels.

For working hours, the dual-exposure group shows more moderate reductions, while the single-exposure group shows sharp declines from higher levels (from approximately 2000 hours to approximately 1400 hours). In terms of physical health, dual-exposure caregivers show more obvious deterioration, possibly related to the continued high-intensity caregiving burden they bear. The biggest difference between the two groups is reflected in income: the single-exposure group shows significant income decline (from approximately 10000 CNY to approximately 8000 CNY), while the dual-exposure group maintains a stable but low-level income (approximately

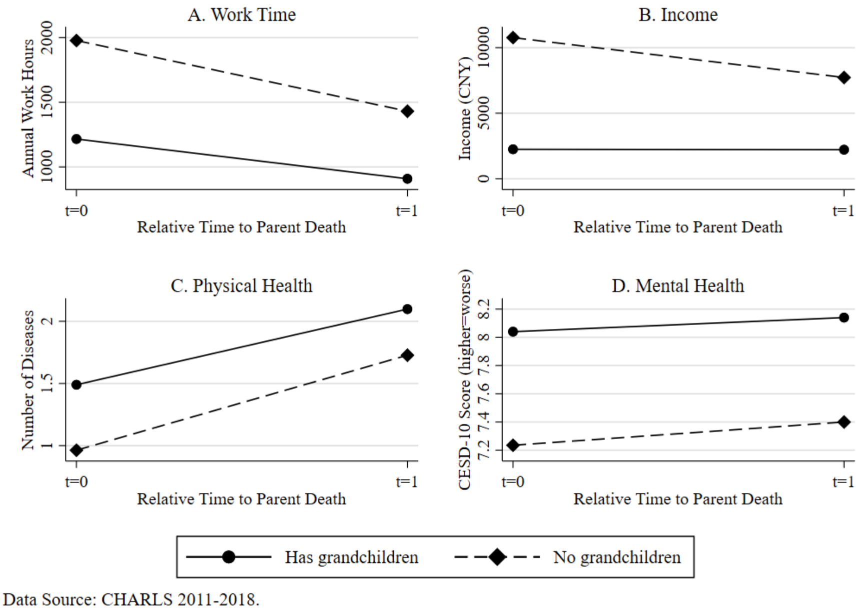

Figure 4: Labor Market and Health Responses to Parent Death Shock

2000 CNY).

Overall, the parental death shock analysis shows the complex patterns of how dual-exposure and single-exposure caregivers respond to exogenous shocks. Both groups show similar change trends, both reduce caregiving time, decrease working hours, and experience health deterioration, but show significant differences in specific response magnitudes and absolute levels. Based on these results, we can preliminarily infer that caregiving type may have certain unique impacts on labor market performance and health outcomes.

4.2.2 Parental Health Shock

Parental health shock is constructed based on the deterioration of parental health status, providing another exogenous shock perspective for analyzing caregiving exposure differences. This study uses respondents' evaluations of the health conditions of four parents (parents and parents-in-law) to construct the shock variable. In the 2011-2013 data, health ratings use a 1-5 scale (1 being the best, 5 being the worst), with levels 1-3 defined as good health and levels 4-5 defined as poor health. Due to the lack of the same health rating data in 2018, this study uses "whether parents have the ability to care for themselves" as a proxy variable for health status. Parental health shock is defined as an event where at least one parent of the respondent transitions from the good health group to the poor health group. Specifically, during 2011-2013, shock identification is based on direct changes in health scores; during 2013-2018, shock identification refers to cases where health scores were good in 2013 but self-care ability obstacles appeared by 2018.

The grouping logic based on caregiving exposure types remains consistent with the parental death shock. The dual-exposure group refers to those who have grandchildren and experience parental health deterioration, facing increased parental care needs while continuing to care for grandchildren. The single-exposure group refers to those without grandchildren but experiencing parental health deterioration, primarily facing increased parental care needs. Under this definition, a total of 713 female samples experienced parental health deterioration during the tracking period. Among those with stable family structure and actual caregiving provision, there are 545 samples, including 353 dual-exposure caregivers (with parental health deterioration + grandchildren) and 192 single-exposure caregivers (with parental health deterioration only).

Unlike the reallocation of caregiving time in parental death shock, the two groups show certain differences in caregiving time adjustments. Figure 5 shows the impact of parental health deterioration shock on caregiving time across different caregiving exposure types. After the shock occurs, both groups relatively reduce caregiving time for parents. This may be because severe parental health deterioration requires respondents to consider professional care rather than self-care, thus objectively reducing caregiving time. From the perspective of relative changes, the group without grandchildren (single-exposure caregivers) significantly reduces parental caregiving time (approximately 72% decline), while the group with grandchildren (dual-exposure caregivers) only slightly reduces it (approximately 8.6%) and simultaneously increases caregiving time for grandchildren, achieving internal reallocation of caregiving resources and a slight increase in total caregiving burden. From the absolute level perspective, the dual-exposure group shows much higher total caregiving time than the single-exposure group, and the gap further increases after the shock, demonstrating the high-intensity caregiving characteristics of this group.

Echoing the caregiving time response patterns, both caregiving exposure groups also show patterns that are both similar and different in welfare outcomes. As shown in Figure 6, in terms of change trends, both groups show similar trends in working hours, income, and physical health: working hours all decline, income all increase, and physical health all deteriorates (increased number of diseases). However, there are significant differences in the magnitude of changes and absolute levels. The biggest difference between the two groups is reflected in

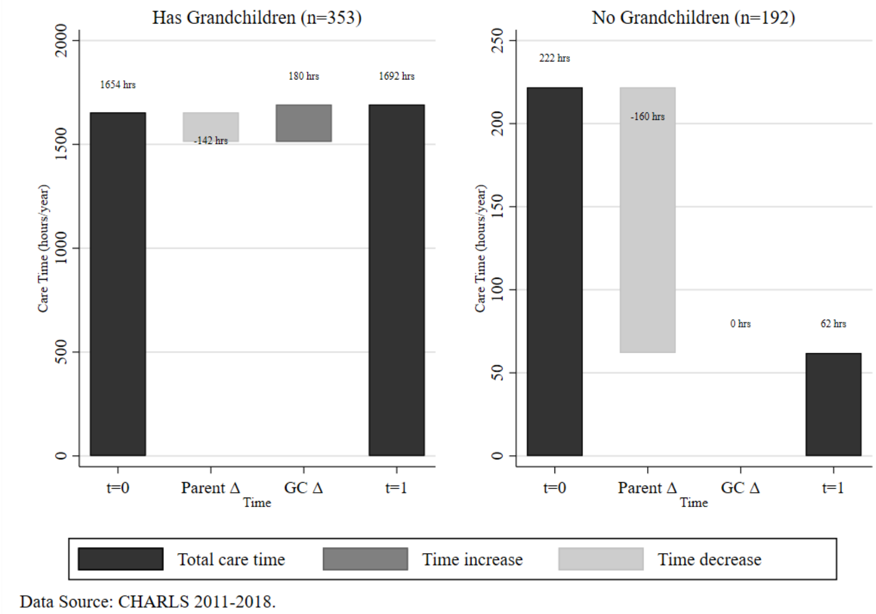

Figure 5: Caregiving Time Reallocation Following Parent Health Shock

mental health: the single-exposure group shows significant mental health deterioration (from approximately 7.0 points to 8.7 points), while the dual-exposure group only shows slight deterioration (from approximately 8.6 points to 8.8 points). However, from the overall absolute level perspective, the dual-exposure group's labor market performance and physical condition are both worse than the single-exposure group, suggesting the potential pressure and impact of dual caregiving.

4.2.3 Grandchild Birth Shock

Grandchild birth shock is defined based on the number of grandchildren aged 16 and below. This study defines grandchild birth shock as an increase in the number of grandchildren aged 16 and below during the sample period. While parental death shock and parental health shock reflect changes in caregiving needs from the parental generation, grandchild birth shock represents positive growth in caregiving needs from the grandchild generation. This shock provides another important perspective for observing the impact of dual caregiving: compared to single-exposure caregivers, when facing new caregiving demands, dual-exposure caregivers not only need to consider directly increasing caregiving investment for grandchildren, but also need to rebalance the allocation of care between parents and grandchildren under limited time and energy constraints. From this perspective, this shock may have more obvious impacts on the labor supply and health status of dual caregivers.

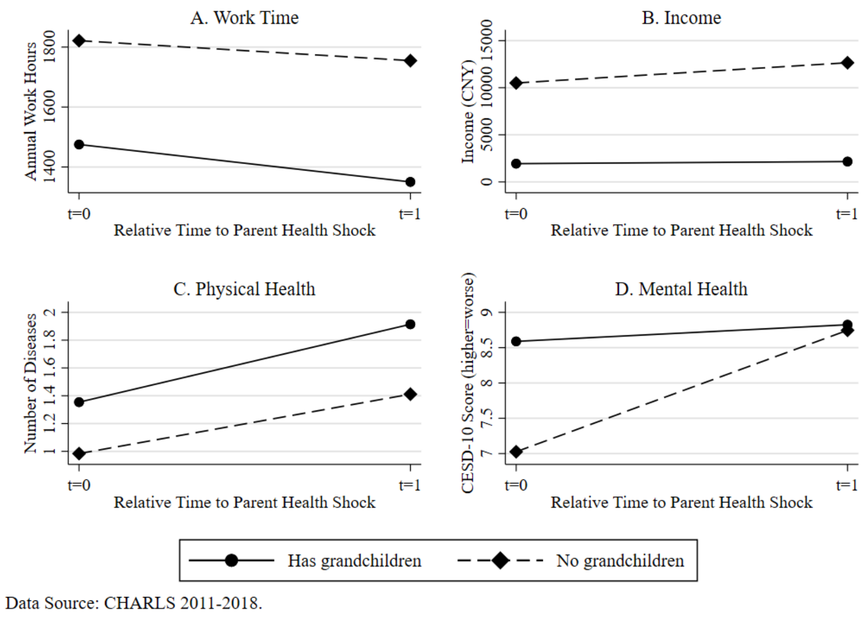

Figure 6: Labor Market and Health Responses to Parent Health Shock

The grouping logic based on caregiving exposure types remains symmetric with the parental health shock: the dual-exposure group refers to those who have unhealthy parents and experience grandchild birth, who need to take on new grandchild caregiving tasks on top of existing parental caregiving burdens; the single-exposure group refers to those without unhealthy parents but experiencing grandchild birth, who primarily face new grandchild caregiving needs. Under this definition, among the 13711 female analysis samples providing grandchild number data, a total of 3870 people experienced grandchild birth shock during the tracking period, accounting for 28.2% of the analysis sample. After screening for stable family structure and stable grandchild caregiving participation, the final analysis sample is 875 people, including 72 dual-exposure caregivers and 803 single-exposure caregivers.

The changes in caregiving time under grandchild birth shock further highlight the high-intensity caregiving of dual-exposure caregivers. Figure 7 shows the impact of grandchild birth shock on caregiving time across different caregiving exposure types. In terms of trends, both dual-exposure and single-exposure caregivers reduce care for parents and increase caregiving time for grandchildren. However, from the perspective of relative changes, the dual-exposure group shows significantly lower increases in total caregiving time compared to the single-exposure group, indicating that the single-exposure group has a stronger caregiving response to grandchild birth. From the absolute values perspective, the caregiving intensity of the dual-exposure group is much higher than the single-exposure group both before and after

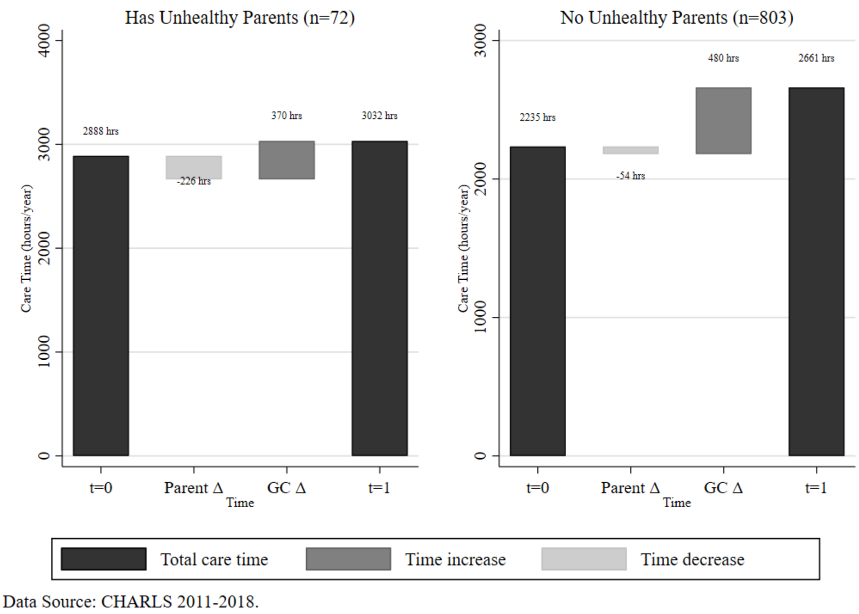

Figure 7: Caregiving Time Reallocation Following Grandchild Birth Shock

the shock, again confirming the high-intensity characteristics of this group.

Grandchild birth shock also produces different impacts on labor and health outcomes across different exposure groups, mainly reflected in mental health. As shown in Figure 8, in terms of change trends, both groups show similar declining trends in working hours and income, with relatively comparable magnitudes of decline. Physical changes also show similar deterioration trends, but the single-exposure group experiences greater health deterioration than the dual-exposure group. The biggest difference between the two groups is reflected in mental health, showing opposite patterns: the dual-exposure group shows significant mental health improvement (from 9.6 points to 8.8 points), while the single-exposure group shows slight deterioration (from 8.9 points to 9.8 points), reflecting the potential impact of caregiving types.

4.2.4 Summary of Shock Analysis

The differential results from the three types of shocks all support the statement that (potential) caregiving types have unique impacts on respondents' labor market performance and health outcomes. However, it should be noted that whether in parental death shock, parental health shock, or grandchild birth shock, dual caregivers all show the highest caregiving time investment, bearing the heaviest caregiving intensity. This is consistent with the results from the static analysis, namely that dual caregiving often accompanies higher caregiving intensity. This phenomenon raises an important identification problem: the differences observed in dual care-

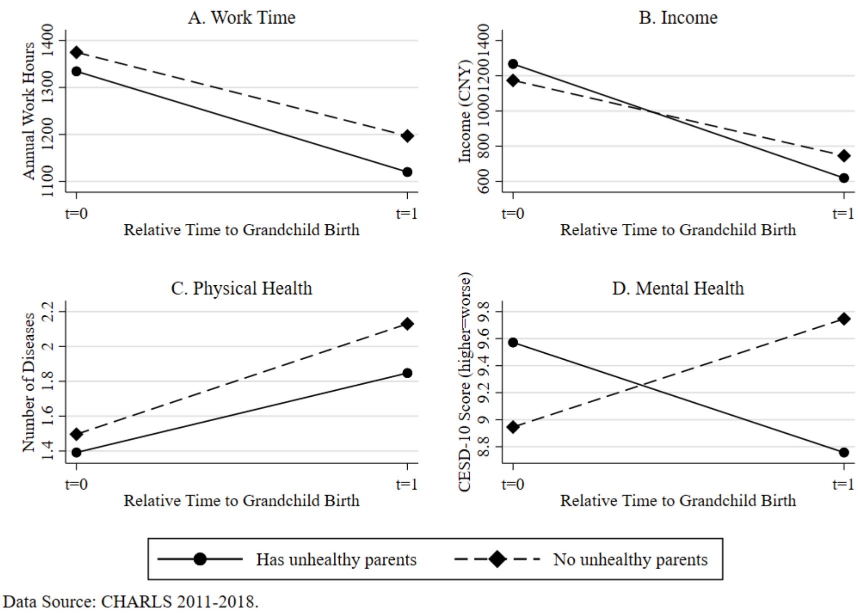

Figure 8: Labor Market and Health Responses to Grandchild Birth Shock

givers' labor participation and health outcomes in the shock analysis may be simultaneously influenced by two factors: first, the uniqueness of caregiving type itself (i.e., the complexity and intergenerational trade-offs brought by simultaneously caring for two generations), and second, the absolute level of caregiving intensity (i.e., the total amount of caregiving time invested). If these two different causal mechanisms cannot be effectively separated, the study will have difficulty determining whether the observed impacts stem entirely from the uniqueness of dual caregiving type or whether caregiving intensity also plays a certain role. To more accurately identify the unique mechanisms of dual caregiving, subsequent analysis will separately examine the independent and interactive effects of caregiving type and caregiving intensity on labor participation and health outcomes.

5 Empirical Strategy and Results

5.1 Empirical Strategy and Model Specification

In the previous analysis, this study used family structure to identify "potential" dual caregivers. The advantage of this definition approach is that it reduces selection bias, because family structure characteristics (particularly exogenous shocks) are largely unaffected by individual subjective decisions. However, there may be significant differences between "potential needs" and "actual behavior": not all women facing caregiving needs will actually undertake caregiving responsibilities. To more precisely identify the direct impact mechanisms of caregiving behavior on individual welfare, the empirical analysis in this section limits the sample to "actual caregivers" - women who explicitly report providing caregiving services in the survey.

Similarly, based on the problem of mixed effects between caregiving type and caregiving intensity identified in the previous analysis, this study constructs the following baseline model to separate their independent effects. The baseline model adopts a two-way fixed effects specification:

$$\begin{aligned} Outcome_{it} = \beta_0 + \beta_1 Dual_{it} + \beta_2 Intensity_{it} + \beta_3 (Dual_{it} \times Intensity_{it}) \\ + \beta_4 X_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \end{aligned} \tag{1}$$

where $Outcome_{it}$ represents individual i's labor market performance (working hours, income) and health outcomes (physical health, mental health) in period t; $Dual_{it}$ is a dummy variable for dual caregiving, taking the value 1 when individuals simultaneously undertake parental care and grandchild care; $Intensity_{it}$ is a dummy variable for caregiving intensity, defined as total caregiving time exceeding the threshold (1040 hours); X_{it} represents individual control variables; α_i and γ_t represent household fixed effects and time fixed effects, respectively.

For the construction of the caregiving intensity variable, this study follows common practices in existing literature by setting 20 hours per week as the high-intensity caregiving threshold. This threshold is widely adopted in caregiving research to distinguish between high-intensity caregiving (20 hours or more per week) and non-high-intensity caregiving (Jacobs et al., 2017; Kolodziej et al., 2022). Therefore, this study converts 20 hours per week to 1040 hours per year as the threshold for high-intensity caregiving.

The core coefficients in the model are β_1, β_2, and β_3. Coefficient β_1 measures the independent effect of dual caregiving type, namely the differential performance of dual caregivers compared to single caregivers under the same caregiving intensity. Specifically, β_1 captures the unique impact brought by simultaneously caring for two generations. If β_1 is significant, it indicates that even under the same caregiving intensity, the complexity of dual caregiving itself will have additional impacts on individuals. Coefficient β_2 measures the independent effect of caregiving intensity, namely the impact of high-intensity caregiving compared to low-intensity caregiving on individual outcomes under the same caregiving type. This coefficient reflects the marginal effect of caregiving time investment, capturing the direct impact of long-term caregiving activities on individual labor participation and health status. Coefficient β_3 measures the interaction effect between dual caregiving and high-intensity caregiving. This coefficient answers an important question: do dual caregivers face additional burdens beyond simple additive effects when undertaking high-intensity caregiving? If β_3 is significant, it indicates that the combination of dual caregiving and high intensity produces a "1+1>2" effect, meaning that the welfare changes of dual caregivers under high-intensity circumstances exceed the simple addition of the dual caregiving effect (β_1) and the high-intensity effect (β_2).

However, as mentioned earlier, the actual caregiver sample suffers from significant selection bias, so this baseline specification faces potential endogeneity problems. First, there may be reverse causality: labor market performance and health status may influence caregiving choices. For example, individuals with poor health and higher income may actively avoid undertaking dual caregiving responsibilities. Second, there are always unobservable individual characteristics in the regression that affect both caregiving decisions and outcome variables, such as personal sense of responsibility and family values, leading to omitted variable bias.

To address these potential confounding issues, this study adopts an instrumental variable strategy to attempt to resolve endogeneity problems. Specifically, the study uses the number of living parents and the number of grandchildren under 16, as well as their interaction term, as instrumental variables for dual caregiving, caregiving intensity, and their interaction term. The specific specification is as follows:

First Stage:

$$\begin{aligned} Dual_{it} &= \pi_0 + \pi_1 NumParents_{it} + \pi_2 NumGrandchildren_{it} \\ &\quad + \pi_3 (NumParents_{it} \times NumGrandchildren_{it}) + \pi_4 X_{it} + \alpha_i + \gamma_t + u_{it} \quad \text{(2a)} \\ Intensity_{it} &= \delta_0 + \delta_1 NumParents_{it} + \delta_2 NumGrandchildren_{it} \\ &\quad + \delta_3 (NumParents_{it} \times NumGrandchildren_{it}) + \delta_4 X_{it} + \alpha_i + \gamma_t + v_{it} \quad \text{(2b)} \\ Dual_{it} \times Intensity_{it} &= \theta_0 + \theta_1 NumParents_{it} + \theta_2 NumGrandchildren_{it} \\ &\quad + \theta_3 (NumParents_{it} \times NumGrandchildren_{it}) + \theta_4 X_{it} + \alpha_i + \gamma_t + w_{it} \quad \text{(2c)} \end{aligned}$$

Second Stage:

$$\begin{aligned} Outcome_{it} &= \beta_0 + \beta_1 \widehat{Dual}_{it} + \beta_2 \widehat{Intensity}_{it} \\ &\quad + \beta_3 (Dual_{it} \widehat{\times Intensity}_{it}) + \beta_4 X_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \end{aligned} \tag{3}$$

These instrumental variables satisfy the three core conditions for valid instrumental variables. First, the relevance condition: the number of living parents directly determines the potential demand for parental care, the number of grandchildren under 16 determines the potential demand for grandchild care, and their interaction term naturally affects whether individuals choose dual caregiving and the level of caregiving intensity, thus having strong correlation with the endogenous explanatory variables. Second, the exogeneity condition: the number of parents and grandchildren is mainly determined by demographic factors, historical paths of family fertility decisions, and the natural evolution of family structure. These factors are largely exogenous to individuals' current labor market performance and health status and will not be reversely affected by current welfare outcomes. Finally, the exclusion restriction: the impact of the number of living parents and grandchildren on individual labor participation and health outcomes mainly operates through the channel of caregiving needs and caregiving arrangements, with no other direct impact pathways, satisfying the requirement that instrumental variables only affect the dependent variable through endogenous variables. Therefore, this set of instrumental variables provides a credible source of exogenous variation for identifying the causal effects of caregiving behavior on individual welfare.

5.2 Descriptive Statistics

Appendix Table A.1 presents the descriptive statistics for the household fixed effects analysis sample. The sample includes 17087 observations from 8240 households, covering complete information for women aged 45-64 across three waves of CHARLS data from 2011, 2013, and 2018.

In terms of caregiving behavior, dual caregivers (simultaneously caring for parents and grandchildren) account for 4%, while high-intensity caregivers (annual caregiving time exceeding 1040 hours) account for 26%. Sample women provide an average of 949.5 hours of total caregiving time per year, but with enormous individual variation (standard deviation of 1765.5 hours), providing an important source of variation for distinguishing the effects of caregiving intensity and caregiving type. Regarding outcome variables, sample women work an average of 1363.4 hours per year, have an average annual income of 2209.3 CNY, suffer from an average of 1.61 chronic diseases, and score an average of 8.77 points on the CESD-10 depression scale. All variables show considerable individual variation, providing sufficient identifying variation for subsequent regression analysis. For instrumental variables, sample women have an average of 0.93 living parents and 0.95 grandchildren under 16, meeting the basic requirements for instrumental variable analysis.

5.3 Baseline Regression Results

In this study, I adopt a stepwise approach of adding control variables and fixed effects to examine the impact of caregiving behavior on labor market and health outcomes. The complete stepwise estimation results of the baseline model are shown in Appendix Table B.1 to Table B.4, with the focus here on discussing the final baseline results after incorporating household fixed effects.

From the perspective of working hours impact, high-intensity caregiving is the main factor affecting women's labor supply. In the most stringent household fixed effects model (Appendix Table B.1, Column 4), high-intensity caregiving significantly reduces women's annual working hours by 95.2 hours, equivalent to approximately 7% of baseline working hours. In contrast, dual caregiving itself has no statistically significant impact on working hours, and the interaction effect between dual caregiving and high-intensity caregiving is also insignificant. Income results show a similar pattern (Appendix Table B.2). High-intensity caregiving leads to a significant reduction in annual income of 627.4 CNY, while the income effect of dual caregiving itself becomes positive but insignificant after incorporating household fixed effects. This result indicates that caregiving intensity rather than caregiving type is the key factor determining labor market losses.

The baseline estimates for health outcomes show differential impacts of caregiving behavior on physical and mental health. Regarding physical health (Appendix Table B.3), after incorporating household fixed effects, all caregiving variable coefficients are insignificant, indicating no significant statistical association between caregiving behavior and the number of chronic diseases. Mental health results (Appendix Table B.4) similarly show that under the most stringent household fixed effects specification, all caregiving variables have no significant impact on CESD-10 depression scale scores. This result may reflect two aspects: first, the impact of caregiving behavior on mental health may involve mutual cancellation of positive and negative effects; second, there may be uncontrolled endogeneity problems that require further identification through instrumental variable methods.

5.4 Instrumental Variable Validity Tests

Appendix Table C.1 presents the first-stage regression results for the instrumental variables, validating the relevance condition of the selected instruments. The results show that the three instrumental variables have significant predictive power for the endogenous explanatory variables: the number of living parents significantly and positively predicts the probability of dual caregiving, which aligns with expectations since an increase in the number of parents directly raises parental care demand. The number of grandchildren under 16 has a weak negative effect on dual caregiving, possibly reflecting that when caregiving resources are limited, increased grandchild care demand may crowd out parental care. The interaction term between the number of parents and grandchildren significantly and positively affects the probability of dual caregiving, indicating that when simultaneously facing caregiving needs from two generations, individuals are more likely to choose dual caregiving. The number of grandchildren under 16 significantly and positively predicts high-intensity caregiving, reflecting the time-intensive characteristics of grandchild care. The interaction term between parents and grandchildren also significantly predicts high-intensity caregiving, confirming that dual caregiving needs often accompany higher caregiving intensity.

The F-statistics for the first-stage regressions are 37.905 (dual caregiving) and 23.095 (high-intensity caregiving), respectively, both far exceeding the empirical threshold of 10, indicating no weak instrument problem. This provides a reliable statistical foundation for subsequent IV estimation.

5.5 Reduced Form Regression Results

The core logic of reduced form regression is to examine whether instrumental variables affect outcome variables by influencing caregiving behavior, rather than directly affecting outcomes through other channels. The specific form is as follows:

$$\begin{aligned} Outcome_{it} = {} & \gamma_0 + \gamma_1 NumParents_{it} + \gamma_2 NumGrandchildren_{it} \\ & + \gamma_3 (NumParents_{it} \times NumGrandchildren_{it}) + \gamma_4 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \end{aligned} \tag{4}$$

Table 1 shows the direct effects of instrumental variables on outcome variables, providing further validation for the effectiveness of the instrumental variables. Regarding labor outcomes, the results show that the number of living parents has no significant direct effect on working hours, but has a positive effect on income. The number of grandchildren under 16 has a significant negative effect on income and a slight negative effect on mental health. The interaction term between the number of parents and grandchildren has a significant negative effect on working hours. Regarding health, the direct effects of instrumental variables on physical health (number of chronic diseases) are all insignificant, indicating that instrumental variables mainly affect physical health through the caregiving channel rather than other mechanisms. Although the impact on mental health is small, it still exists.

Table 1: Reduced Form Results

	Work Time	Income	Physical Health	Mental Health
Number of Living Parents	45.996	222.167**	-0.008	0.271**
	(33.820)	(111.700)	(0.020)	(0.129)
Number of Grandchildren	-0.124	-156.381***	0.009	0.145**
	(13.080)	(31.320)	(0.009)	(0.057)
Living Parents × Grandchildren	-30.017***	6.061	-0.009	0.009
	(11.413)	(32.585)	(0.008)	(0.049)
Observations	7723	14626	14457	13557
R-squared	0.627	0.708	0.880	0.678

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.6 Main Results

Table 2 presents the impact of caregiving behavior on labor market outcomes. To address the endogeneity problem of caregiving decisions, this study reports both baseline fixed effects estimates and instrumental variable estimates.

Regarding working hours, caregiving behavior shows negative effects on women's labor supply. The instrumental variable estimation results show that although all coefficients are statistically insignificant, they are still worth attention from an economic significance perspective. High-intensity caregiving reduces women's annual working hours by 414.2 hours, equivalent to approximately 8 hours less work per week, reflecting the substantial constraints that caregiving responsibilities impose on labor participation. The independent effect of dual caregiving itself is -259.5 hours, also statistically insignificant, but the magnitude indicates it has a certain negative impact on working hours.

The impact of caregiving behavior on income presents a more complex pattern. The instru-

Table 2: Labor Outcomes: Baseline vs IV Estimates

	Baseline (Household+Year FE)		IV (Household+Year FE)	
	Work Time	Income	Work Time	Income
Dual Caregiving	-51.202	478.502		
	(116.858)	(372.366)		
High Intensity	-95.234***	-627.389***		
	(35.854)	(102.925)		
Dual × Intensity	-19.763	-456.734		
	(142.414)	(447.363)		
Dual $\widehat{\text{Caregiving}}$			-259.465	4551.714**
			(622.594)	(1907.622)
High $\widehat{\text{Intensity}}$			-414.198	-4709.876***
			(351.485)	(873.283)
Dual × $\widehat{\text{Intensity}}$			392.550	-247.002
			(895.337)	(2574.848)
Observations	7723	14626	7723	14626
R-squared	0.627	0.709	0.626	0.708

Notes: Comparison of baseline and IV estimates for labor outcomes. Physical health measured as number of diseases (0-14), mental health measured by CESD-10 score (higher=worse). All regressions include household and time fixed effects with robust standard errors. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics: Dual = 37.91, Intensity = 23.09. $\widehat{}$ indicates IV estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

mental variable estimation results show that high-intensity caregiving has a significant negative impact on income, reducing annual income by 4709.9 CNY. Meanwhile, dual caregiving itself has a significant positive impact on income, significantly increasing the sample's annual income by 4551.7 CNY. The interaction effect between dual caregiving and high-intensity caregiving is statistically insignificant, indicating that dual caregiving does not further amplify the negative economic effects of high-intensity caregiving.

Table 3 presents the impact of caregiving behavior on health outcomes.

Regarding the number of chronic diseases, instrumental variable estimates show that all caregiving variable coefficients are insignificant. This indicates that within the observation period of this study (3-7 years), the impact of caregiving behavior on physical health has not fully manifested, which may be related to the long-term characteristics of chronic disease development.

The impact of caregiving behavior on mental health is more apparent. Instrumental variable estimation results show that high-intensity caregiving significantly increases the CESD-10 depression scale score by 3.16 points, indicating that long-term caregiving activities indeed have negative impacts on caregivers' mental health. The coefficient for dual caregiving itself on men-

Table 3: Health Market Outcomes: Baseline vs IV Estimates

	Baseline (Household+Year FE)		IV (Household+Year FE)	
	Physical Health	Mental Health	Physical Health	Mental Health
Dual Caregiving	-0.044	-0.180		
	(0.073)	(0.519)		
High Intensity	0.017	0.171		
	(0.021)	(0.141)		
Dual × Intensity	-0.008	0.057		
	(0.084)	(0.605)		
Dual $\widehat{\text{Caregiving}}$			-0.235	3.241
			(0.398)	(2.513)
High $\widehat{\text{Intensity}}$			0.109	3.161**
			(0.236)	(1.463)
Dual $\times \widehat{\text{Intensity}}$			-0.169	-1.532
			(0.587)	(3.505)
Observations	14457	13557	14457	13557
R-squared	0.880	0.678	0.880	0.678

Notes: Comparison of baseline and IV estimates for health outcomes. Physical health measured as number of diseases (0-14), mental health measured by CESD-10 score (higher=worse). All regressions include household and time fixed effects with robust standard errors. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics: Dual = 37.91, Intensity = 23.09. $\widehat{}$ indicates IV estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tal health is positive (3.24 points) but statistically insignificant, suggesting that simultaneously caring for two generations may generate some psychological pressure, but the statistical reliability of this effect is limited. The interaction effect coefficient between dual caregiving and high-intensity caregiving is negative (-1.53 points) but also insignificant.

Based on this, in the main analysis, the study has the following three findings: First, caregiving intensity is more important than caregiving type. Across various outcomes, high-intensity caregiving often shows larger impacts (whether significant or not), while the independent effect of dual caregiving is relatively limited. Second, caregiving has differential impacts on different outcomes. The impact of caregiving on income and mental health is more significant, while the impact on working hours and physical health, although in the expected direction, has limited statistical significance. Third, the effects of dual caregiving are complex. Although dual caregiving may bring pressure in terms of mental health, it shows positive effects economically, which may reflect complex resource allocation mechanisms within families or large heterogeneity differences among samples.

6 Robustness Checks and Heterogeneity Analysis

6.1 Robustness Checks

To ensure the reliability of the main conclusions, this study conducts robustness checks by changing the caregiving intensity threshold. Specifically, the baseline analysis uses 1040 hours/year (20 hours per week) as the threshold for high-intensity caregiving. To test the sensitivity of results to threshold selection, 520 hours/year (10 hours per week) and 2080 hours/year (40 hours per week) are used as alternative thresholds for robustness checks. When using the 520-hour threshold, the proportion of high-intensity caregivers increases to 31%; when using the 2080-hour threshold, this proportion decreases to 17%. These distribution changes provide a foundation for testing the consistency of results under different intensity definitions.

Overall, the main findings of this study show robustness to the selection of caregiving intensity thresholds, with the negative effects of high-intensity caregiving and the complex effects of dual caregiving remaining consistent under different threshold definitions. As shown in the 520-hour threshold results in Appendix Table D.1 and D.3, under the lower intensity threshold, the negative impact of high-intensity caregiving on income remains significant (-4017.5 CNY, $p<0.01$), and the negative impact on mental health is also significant (2.82 points, $p<0.05$). The positive income effect of dual caregiving remains robust (4273.7 CNY, $p<0.05$). The 2080-hour threshold results shown in Appendix Table D.2 and D.4 demonstrate that under the more stringent intensity threshold, the negative effects of high-intensity caregiving are further amplified. Income loss significantly increases to -5176.8 CNY, and mental health deterioration is 3.15 points. The positive income effect of dual caregiving also remains significant (3635.2 CNY, $p<0.05$). In conclusion, the results and conclusions can be considered robust across various threshold intensities.

6.2 Heterogeneity Analysis

To gain a deeper understanding of the differential characteristics of caregiving behavior impacts, this study conducts heterogeneity analysis from two dimensions: education level and urban-rural areas.

6.2.1 Educational Heterogeneity Analysis

The sample is divided into three groups by education level: low education (below junior high school), medium education (high school and vocational education), and high education (college and above). The analysis results show (see Appendix Tables E.1 to E.4 for details) that the impacts of caregiving behavior differ significantly across education groups. In the low education group, dual caregiving responsibilities slightly increase working hours and income, but high-intensity caregiving still has a significant negative impact on income while reducing working time. Notably, the interaction term between dual caregiving and high-intensity caregiving shows a significant negative impact on working time, indicating that when low-educated women simultaneously undertake dual high-intensity caregiving, their labor participation is severely squeezed. In terms of health, the negative impact of high-intensity caregiving on mental health is particularly significant for the low education group. In contrast, the medium and high education groups have limited statistical power in their estimation results due to smaller sample sizes. These findings suggest that women with lower education levels face greater economic and health risks when undertaking caregiving responsibilities.

6.2.2 Urban-Rural Heterogeneity Analysis

More significant heterogeneity appears in the differences between urban and rural samples. Appendix Table E.5 shows a comparison of basic characteristics between rural and urban samples. Urban women have longer annual working hours, and their average annual income (3091.8 CNY) is 2.5 times that of rural women (1226.3 CNY), with significantly higher education levels, but relatively better mental health status (depression score 7.73 vs 9.57). In terms of caregiving behavior, urban-rural differences are relatively limited: dual caregiving proportions are 3.5% (rural) and 4.3% (urban) respectively.

The instrumental variable estimation results in Appendix Table E.7 and E.8 show important urban-rural differences in caregiving behavior impacts. Labor market results show that urban women bear more severe economic shocks. The negative impact of high-intensity caregiving on urban women's income (-4767.7 CNY, $p<0.01$) is approximately 2.2 times that of rural women (-2190.7 CNY, $p<0.05$). Meanwhile, dual caregiving has a positive impact on working time in the rural sample (277.3 hours, $p<0.01$), reflecting more flexible labor arrangements in rural areas. In terms of health, both show similar results: the negative impact of high-intensity caregiving on mental health is significant and of similar magnitude in both urban and rural areas (rural: 4.78 points, urban: 4.13 points).

This urban-rural difference reflects essential differences in the main challenges faced by women in different regions. Urban labor markets have a higher degree of formalization, making time conflicts between caregiving responsibilities and formal employment more severe, leading to greater income losses. Although rural areas have relatively flexible labor arrangements and women can more easily find balance between caregiving and work, the relatively backward economic development level results in lower absolute income levels for caregivers.

7 Conclusion

Based on data from the China Health and Retirement Longitudinal Study (CHARLS), this study employs a progressive research design incorporating static analysis, dynamic shock analysis, and instrumental variable causal identification to systematically analyze the impact of dual caregiving behavior on middle-aged women's labor supply and health status. The study first discovers through static and shock analysis of potential caregivers that dual caregiving exposure individuals indeed bear higher caregiving time burdens and more complex caregiving situations, exhibiting different response patterns from single caregivers when facing exogenous shocks such as parental death, parental health deterioration, and grandchild birth, providing an important descriptive evidence foundation for subsequent causal analysis.

The instrumental variable analysis based on actual caregiving behavior reveals the core mechanisms of caregiving impact. The study finds that caregiving intensity rather than caregiving type is the key factor determining caregiving impacts. High-intensity caregiving has comprehensive negative effects on women's labor market performance and mental health, leading to significant income losses and mental health deterioration. In contrast, the independent effect of dual caregiving itself is relatively limited, with most effects being statistically insignificant. Notably, dual caregiving has a significant positive impact on women's income, but the specific reasons for this still require further research. The interaction effects between dual caregiving and high-intensity caregiving are insignificant across all outcomes, indicating that dual caregiving does not further amplify the negative effects of high-intensity caregiving.

Heterogeneity analysis shows important group differences in caregiving impacts, with urban-rural differences being particularly significant. Urban women bear economic shocks approximately twice those of rural women, reflecting differentiated characteristics of caregiving conflicts under different labor market structures. Urban labor markets have a higher degree of formalization, making time conflicts between caregiving responsibilities and formal employment more severe, while rural areas have relatively flexible labor arrangements, making it easier for women to find balance between caregiving and work. Educational level analysis shows that low-education groups face greater economic and health risks when undertaking caregiving responsibilities.

The theoretical contributions of this study are demonstrated in multiple aspects. In terms of research content, this study is the first to systematically incorporate dual caregiving into the analytical framework and effectively separate the different roles of caregiving type and caregiving intensity, finding that caregiving intensity is the key factor affecting caregiving consequences. Methodologically, by constructing instrumental variable strategies using the number of living parents, number of grandchildren, and their interaction term, the study attempts to address endogeneity problems in dual caregiving research and achieve effective causal identification. Finally, the study adopts a progressive analytical design from potential caregiving exposure to actual caregiving behavior, providing a new analytical framework for caregiving research.

The limitations of the study are mainly shown in the following aspects. In terms of data, the study is limited by the survey cycle of CHARLS data, mainly focusing on short to medium-term impacts, and some key variables are missing, making it impossible to fully capture the long-term cumulative effects of caregiving behavior. Methodologically, although the instrumental variable strategy largely reduces endogeneity problems, the strict exogeneity assumption of current instrumental variables still needs further verification, and the current linear model specification may not fully capture the nonlinear characteristics of caregiving impacts. Accordingly, further extensions could consider: extending the observation period to analyze the long-term effects of caregiving behavior; exploring in depth the theoretical mechanisms behind the complex effects of dual caregiving; introducing more heterogeneity dimensions and nonlinear models to comprehensively understand the complex characteristics of caregiving impacts.

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A Summary Statistics

Table A.1: Descriptive Statistics - Main Regression

	Mean	Std. Dev.	Min	Max	N
Work Time (hours/year)	1363.38	1052.49	0.00	3345.65	10461
Income (CNY)	2209.25	5556.70	0.00	20000.00	16918
Physical Health (disease count)	1.61	1.49	0.00	5.00	16785
Mental Health (CESD-10)	8.77	6.09	0.00	22.00	15976
Dual Caregiving (binary)	0.04	0.19	0.00	1.00	17087
High Intensity Care (binary)	0.26	0.44	0.00	1.00	17087
Parent Care Hours	54.46	245.38	0.00	1680.00	17087
Grandchild Care Hours	895.05	1745.11	0.00	7280.00	17087
Total Care Hours	949.50	1765.52	0.00	8960.00	17087
Number of Living Parents	0.93	1.08	0.00	4.00	17087
Number of Grandchildren	0.95	1.72	0.00	35.00	17087
Parents × Grandchildren	0.66	1.90	0.00	33.00	17087
Age	55.18	6.96	38.00	72.00	17087

Notes: Summary statistics for female analysis sample using waves 1, 2, and 4 (2011, 2013, 2018). Physical health measured as number of diseases (0-14), mental health measured by CESD-10 score (higher=worse). High intensity defined as total care hours above 1040 hours per year (20 hours per week).

B Additional Tables for Baseline Regression

Table B.1: Labor Market Outcomes: Work Time - Stepwise Controls

	(1)	(2)	(3)	(4)
Dual Caregiving	-56.213	-82.987	-126.250	-51.202
	(101.251)	(101.369)	(101.064)	(116.858)
High Intensity	-277.843***	-232.711***	-223.392***	-95.234***
	(24.740)	(24.833)	(24.727)	(35.854)
Dual × Intensity	159.321	106.037	112.227	-19.763
	(120.778)	(121.492)	(120.694)	(142.414)
Controls	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Household FE	No	No	No	Yes
Observations	10461	10461	10461	7723
R-squared	0.012	0.033	0.039	0.627

Notes: Stepwise addition of controls and fixed effects for work time outcomes. High intensity defined as total care hours above 1040 hours per year. Controls include age. All standard errors are robust. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Labor Market Outcomes: Income - Stepwise Controls

	(1)	(2)	(3)	(4)
Dual Caregiving	-721.572*	-958.481**	-670.957*	478.502
	(389.699)	(385.426)	(386.791)	(372.366)
High Intensity	-1548.523***	-1235.840***	-1337.957***	-627.389***
	(81.200)	(78.525)	(79.717)	(102.925)
Dual × Intensity	1356.778***	949.853**	925.153**	-456.734
	(458.086)	(454.042)	(455.401)	(447.363)
Controls	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Household FE	No	No	No	Yes
Observations	16918	16918	16918	14626
R-squared	0.014	0.065	0.075	0.709

Notes: Stepwise addition of controls and fixed effects for income outcomes. High intensity defined as total care hours above 1040 hours per year. Controls include age. All standard errors are robust. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3: Health Outcomes: Physical Health - Stepwise Controls

	(1)	(2)	(3)	(4)
Dual Caregiving	-0.146	-0.067	0.041	-0.044
	(0.114)	(0.114)	(0.115)	(0.073)
High Intensity	0.176***	0.075***	0.042	0.017
	(0.028)	(0.027)	(0.027)	(0.021)
Dual × Intensity	-0.174	-0.040	-0.044	-0.008
	(0.134)	(0.134)	(0.133)	(0.084)
Controls	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Household FE	No	No	No	Yes
Observations	16785	16785	16785	14457
R-squared	0.003	0.081	0.100	0.880

Notes: Stepwise addition of controls and fixed effects for physical health outcomes (disease count). High intensity defined as total care hours above 1040 hours per year. Controls include age. All standard errors are robust. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Health Outcomes: Mental Health - Stepwise Controls

	(1)	(2)	(3)	(4)
Dual Caregiving	-0.486	-0.391	-0.349	-0.180
	(0.496)	(0.498)	(0.497)	(0.519)
High Intensity	0.310***	0.182	0.189*	0.171
	(0.114)	(0.115)	(0.115)	(0.141)
Dual × Intensity	-0.343	-0.174	-0.159	0.057
	(0.578)	(0.580)	(0.578)	(0.605)
Controls	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Household FE	No	No	No	Yes
Observations	15976	15976	15976	13557
R-squared	0.001	0.008	0.010	0.678

Notes: Stepwise addition of controls and fixed effects for mental health outcomes (CESD-10 score). High intensity defined as total care hours above 1040 hours per year. Controls include age. All standard errors are robust. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C First Stage IV Results

Table C.1: First Stage Results - Main Regression

	Dual Caregiving	High Intensity
Number of Living Parents	0.044***	0.003
	(0.005)	(0.011)
Number of Grandchildren	-0.002*	0.021***
	(0.001)	(0.004)
Parents × Grandchildren	0.011***	0.009**
	(0.002)	(0.004)
F-statistic	37.905	23.095
Observations	14786	14786
R-squared	0.504	0.568

Notes: First stage regressions with household and time fixed effects and robust standard errors. High intensity defined as total care hours above 1040 hours per year. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics test joint significance of all instruments. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Robustness Checks

Table D.1: Robustness Check 1: First Stage Results - 520 Hour Threshold

	Dual Caregiving	High Intensity (520h)
Number of Living Parents	0.044***	0.008
	(0.005)	(0.011)
Number of Grandchildren	-0.002*	0.026***
	(0.001)	(0.004)
Parents × Grandchildren	0.011***	0.006*
	(0.002)	(0.004)
F-statistic	37.905	30.128
Observations	14786	14786
R-squared	0.504	0.587

Notes: First stage regressions for 520 hour intensity threshold (10 hours/week). All regressions include household and time fixed effects with robust standard errors. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics test joint significance of all instruments. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.2: Robustness Check 2: First Stage Results - 2080 Hour Threshold

	Dual Caregiving	High Intensity (2080h)
Number of Living Parents	0.044***	-0.017*
	(0.005)	(0.009)
Number of Grandchildren	-0.002*	0.017***
	(0.001)	(0.003)
Parents × Grandchildren	0.011***	0.008**
	(0.002)	(0.003)
F-statistic	37.905	20.196
Observations	14786	14786
R-squared	0.504	0.534

Notes: First stage regressions for 2080 hour intensity threshold (40 hours/week). All regressions include household and time fixed effects with robust standard errors. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics test joint significance of all instruments. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.3: Robustness Check 1: 520 Hour Threshold

	Baseline (Household+Year FE)				IV (Household+Year FE)			
	Work Time	Income	Physical	Mental	Work Time	Income	Physical	Mental
Dual Caregiving	-204.675	1030.511*	-0.071	-0.539				
	(135.083)	(564.229)	(0.094)	(0.657)				
High Intensity (520h)	-72.670**	-619.328***	0.001	0.159				
	(33.905)	(100.254)	(0.020)	(0.134)				
Dual × Intensity	162.267	-1062.224*	0.031	0.489				
	(156.888)	(603.921)	(0.102)	(0.722)				
Dual $\widehat{\text{Caregiving}}$					-257.809	4273.679**	-0.365	2.370
					(647.295)	(2002.966)	(0.422)	(2.580)
High $\widehat{\text{Intensity}}$ (520h)					-277.072	-4017.455***	0.073	2.817**
					(304.320)	(742.107)	(0.203)	(1.275)
Dual × $\widehat{\text{Intensity}}$					163.506	-347.922	0.166	0.813
					(858.726)	(2475.323)	(0.564)	(3.385)
Observations	7723	14626	14457	13557	7723	14626	14457	13557
R-squared	0.627	0.709	0.880	0.678	0.626	0.708	0.880	0.678

Notes: Robustness check using 520 hour intensity threshold (10 hours/week). All regressions include household and time fixed effects with robust standard errors. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics: Dual = 37.91, Intensity = 30.13. $\hat{}$ indicates IV estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.4: Robustness Check 2: 2080 Hour Threshold

	Baseline (Household+Year FE)				IV (Household+Year FE)			
	Work Time	Income	Physical	Mental	Work Time	Income	Physical	Mental
Dual Caregiving	-43.076	50.252	-0.053	-0.035				
	(94.289)	(320.647)	(0.052)	(0.378)				
High Intensity (2080h)	-89.366**	-526.876***	0.023	0.111				
	(41.915)	(105.860)	(0.024)	(0.156)				
Dual × Intensity	-64.282	33.880	0.004	-0.138				
	(138.760)	(434.890)	(0.074)	(0.528)				
Dual $\widehat{\text{Caregiving}}$					-227.456	3635.213**	-0.248	4.206*
					(559.518)	(1797.590)	(0.364)	(2.252)
High $\widehat{\text{Intensity}}$ (2080h)					-558.139	-5176.789***	0.115	3.153*
					(403.503)	(1040.263)	(0.268)	(1.649)
Dual $\widehat{\times\ \text{Intensity}}$					174.573	-2132.989	-0.089	-2.175
					(1111.596)	(3230.679)	(0.706)	(4.272)
Observations	7723	14626	14457	13557	7723	14626	14457	13557
R-squared	0.627	0.708	0.880	0.678	0.626	0.708	0.880	0.678

Notes: Robustness check using 2080 hour intensity threshold (40 hours/week). All regressions include household and time fixed effects with robust standard errors. Controls include age. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics: Dual = 37.91, Intensity = 20.20. $\hat{}$ indicates IV estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Heterogeneity Analysis

E.1 Education Heterogeneity Analysis

Table E.1: Heterogeneity Analysis - Education Level Comparison

	Low Education (< Lower Secondary)	Mid Education (Upper Secondary)	High Education (Tertiary)
Outcome Variables			
Work Time (hours/year)	1337.85	1631.62	1758.98
Income (CNY)	1790.65	5329.45	14020.83
Physical Health (disease count)	1.63	1.46	1.17
Mental Health (CESD-10)	9.03	6.33	4.84
Treatment Variables			
Dual Caregiving (binary)	0.036	0.042	0.033
High Intensity 1040h (binary)	0.262	0.222	0.154
Total Care Hours	973.04	765.79	416.58
Parent Care Hours	48.25	106.25	172.40
Grandchild Care Hours	924.78	659.55	244.18
Observations	15463	1439	182

Notes: Comparison of means across education levels for heterogeneity analysis. Education levels: Low (less than lower secondary), Mid (upper secondary & vocational training), High (tertiary education).

Table E.2: First Stage Results by Education Level

	Low Education		Mid Education		High Education	
	Dual	Intensity	Dual	Intensity	Dual	Intensity
Number of Living Parents	0.045***	-0.002	0.046***	0.068**	-0.146	-0.120
	(0.006)	(0.011)	(0.016)	(0.031)	(0.106)	(0.154)
Number of Grandchildren	-0.002*	0.019***	0.000	0.049***	-0.027	-0.056
	(0.001)	(0.004)	(0.005)	(0.014)	(0.028)	(0.162)
Parents × Grandchildren	0.010***	0.010***	0.033***	-0.014	0.208	0.145
	(0.002)	(0.004)	(0.011)	(0.016)	(0.162)	(0.183)
F-statistic	33.571	20.554	7.323	5.407	1.018	.339
Observations	13512	13512	1161	1161	111	111
R-squared	0.498	0.566	0.558	0.598	0.800	0.686

Notes: First stage regressions with household and time fixed effects by education level using 1040 hour intensity threshold (20 hours/week). Low education: less than lower secondary (15463 obs, 7359 households). Mid education: upper secondary & vocational (1439 obs, 758 households). High education: tertiary (182 obs, 121 households). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.3: Labor Market Outcomes by Education Level

	Low Education				Mid Education		High Education	
	Baseline		IV		Baseline		Baseline	
	Work	Income	Work	Income	Work	Income	Work	Income
Dual Caregiving	-49.071	362.001	204.299**	805.653***	-223.078	1895.379	0.000	-6926.985
	(119.514)	(326.635)	(89.233)	(245.591)	(203.163)	(2770.964)	(.)	(6457.323)
High Intensity (1040h)	-94.635**	-588.165***	-554.326*	-3487.370***	-136.561	-1176.938*	236.104	690.815
	(37.004)	(101.475)	(326.769)	(826.653)	(154.254)	(604.876)	(197.294)	(1566.540)
Dual × Intensity	-21.467	-397.891	-279.749*	-604.608	156.872	-266.937	0.000	0.000
	(147.764)	(411.315)	(169.015)	(450.224)	(234.090)	(2910.418)	(.)	(.)
Observations	7186	13387	7186	13387	475	1130	62	107
R-squared	0.623	0.661	0.623	0.661	0.670	0.780	0.636	0.863

Notes: Labor market outcomes by education level using 1040 hour intensity threshold (20 hours/week). All regressions include household and time fixed effects with robust standard errors. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics: Low Edu (Dual = 33.57, Intensity = 20.55), Mid Edu (Dual = 7.32, Intensity = 5.41), High Edu (Dual = 1.02, Intensity = 0.34). IV estimates are reported only for Low Education group where instruments are strong. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.4: Health Outcomes by Education Level

	Low Education				Mid Education		High Education	
	Baseline		IV		Baseline		Baseline	
	Physical	Mental	Physical	Mental	Physical	Mental	Physical	Mental
Dual Caregiving	-0.034	-0.063	-0.019	-0.197	-0.176	-1.973	-0.075	8.307***
	(0.075)	(0.532)	(0.054)	(0.357)	(0.288)	(2.076)	(0.261)	(3.041)
High Intensity (1040h)	0.017	0.278*	-0.028	3.751***	0.021	-1.082**	0.180	-5.101*
	(0.022)	(0.147)	(0.229)	(1.396)	(0.075)	(0.515)	(0.154)	(2.899)
Dual × Intensity	-0.013	-0.251	-0.003	-0.082	0.047	3.737*	0.000	0.000
	(0.087)	(0.625)	(0.100)	(0.676)	(0.334)	(2.261)	(.)	(.)
Observations	13218	12424	13218	12424	1131	1043	106	88
R-squared	0.879	0.674	0.879	0.674	0.892	0.670	0.953	0.667

Notes: Health outcomes by education level using 1040 hour intensity threshold (20 hours/week). All regressions include household and time fixed effects with robust standard errors. Instruments include number of living parents, number of grandchildren, and their interaction. F-statistics: Low Edu (Dual = 33.57, Intensity = 20.55), Mid Edu (Dual = 7.32, Intensity = 5.41), High Edu (Dual = 1.02, Intensity = 0.34). IV estimates are reported only for the Low Education group where instruments are strong. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.2 Urban-Rural Heterogeneity Analysis

Table E.5: Heterogeneity Analysis - Rural vs Urban Comparison

	Rural	Urban
Outcome Variables		
Work Time (hours/year)	1273.41	1502.21
Income (CNY)	1226.29	3091.83
Physical Health (disease count)	1.63	1.66
Mental Health (CESD-10)	9.57	7.73
Treatment Variables		
Dual Caregiving (binary)	0.035	0.043
High Intensity 1040h (binary)	0.256	0.271
Total Care Hours	949.45	977.71
Parent Care Hours	46.89	70.36
Grandchild Care Hours	902.56	907.35
Observations	8934	5539

Notes: Comparison of means between rural and urban subsamples for heterogeneity analysis. Rural status determined by baseline household registration. Physical health measured as number of diseases (0-14), mental health measured by CESD-10 score (higher=worse). High intensity threshold: 1040h = 20 hours/week. Sample includes waves 1, 2, and 4 (2011, 2013, 2018).

Table E.6: First Stage Results by Rural/Urban Status

	Rural		Urban	
	Dual	High Intensity	Dual	High Intensity
Number of Living Parents	0.048***	0.010	0.044***	0.002
	(0.008)	(0.014)	(0.009)	(0.018)
Number of Grandchildren	-0.002	0.015***	-0.002	0.037***
	(0.001)	(0.004)	(0.002)	(0.008)
Parents × Grandchildren	0.012***	0.011**	0.010**	0.004
	(0.003)	(0.005)	(0.004)	(0.008)
F-statistic	26.312	11.887	11.359	10.842
Observations	8511	8511	5053	5053
R-squared	0.489	0.555	0.527	0.583

Notes: First stage regressions with household and time fixed effects by rural/urban status using 1040 hour intensity threshold (20 hours/week). Instruments include number of living parents, number of grandchildren, and their interaction. Controls include age, married, and rural. F-statistics test joint significance of all instruments. Rural sample: 8934 obs, 3728 households. Urban sample: 5539 obs, 2513 households. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.7: Labor Market Outcomes by Rural/Urban Status

	Rural				Urban			
	Baseline		IV		Baseline		IV	
	Work Time	Income	Work Time	Income	Work Time	Income	Work Time	Income
Dual Caregiving	-21.997	520.483			-198.750	-10.744		
	(132.088)	(385.219)			(239.459)	(640.264)		
High Intensity (1040h)	-117.535***	-281.740**			-101.602	-1019.561***		
	(42.362)	(115.172)			(76.366)	(203.918)		
Dual × Intensity	-56.177	-404.342			185.783	132.789		
	(164.843)	(460.811)			(282.810)	(807.911)		
Dual $\widehat{\text{Caregiving}}$			277.320***	413.958			-69.847	624.781
			(99.202)	(284.372)			(174.991)	(483.631)
High $\widehat{\text{Intensity}}$ (1040h)			-320.406	-2190.697**			-552.735	-4767.699***
			(390.856)	(973.376)			(579.758)	(1349.176)
Dual $\widehat{\times \text{Intensity}}$			-380.570**	-280.349			268.322	379.886
			(188.785)	(527.573)			(321.616)	(892.707)
Observations	5317	8453	5317	8453	1851	4983	1851	4983
R-squared	0.607	0.637	0.607	0.637	0.645	0.748	0.645	0.748

Notes: Labor market outcomes by rural/urban status using 1040 hour intensity threshold (20 hours/week). All regressions include household and time fixed effects with robust standard errors. Instruments include number of living parents, number of grandchildren, and their interaction. Controls include age, married, and rural. Rural F-stats: Dual = 26.31, Intensity = 11.89. Urban F-stats: Dual = 11.36, Intensity = 10.84. $\hat{}$ indicates IV estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.8: Health Outcomes by Rural/Urban Status

	Rural				Urban			
	Baseline		IV		Baseline		IV	
	Physical	Mental	Physical	Mental	Physical	Mental	Physical	Mental
Dual Caregiving	-0.060	-0.032			-0.033	-0.496		
	(0.094)	(0.756)			(0.104)	(0.648)		
High Intensity (1040h)	0.021	0.151			0.002	0.347		
	(0.027)	(0.190)			(0.034)	(0.228)		
Dual × Intensity	0.024	0.382			-0.012	-0.413		
	(0.108)	(0.863)			(0.125)	(0.802)		
Dual $\widehat{\text{Caregiving}}$			-0.047	-0.340			-0.028	-0.123
			(0.066)	(0.456)			(0.078)	(0.520)
High $\widehat{\text{Intensity}}$ (1040h)			-0.383	4.783***			0.299	4.134**
			(0.312)	(1.801)			(0.269)	(1.702)
Dual $\widehat{\times \text{Intensity}}$			0.044	0.518			0.049	-0.437
			(0.123)	(0.863)			(0.149)	(0.989)
Observations	8448	7884	8448	7884	5005	4611	5005	4611
R-squared	0.874	0.663	0.874	0.664	0.895	0.678	0.896	0.679

Notes: Health outcomes by rural/urban status using 1040 hour intensity threshold (20 hours/week). All regressions include household and time fixed effects with robust standard errors. Instruments include number of living parents, number of grandchildren, and their interaction. Controls include age, married, and rural. Rural F-stats: Dual = 26.31, Intensity = 11.89. Urban F-stats: Dual = 11.36, Intensity = 10.84. $\hat{}$ indicates IV estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.